

S1 Data Description

S1.1 Peer Review Data

Our primary dataset comprises comprehensive peer review records from a single journal published by a leading medical research publisher, spanning a ten-year period from 2012 to 2022. It comprises approximately 37K manuscript submissions and 551K associated review records. Each record captures the full review invitation process, including rich manuscript metadata—such as title, abstract, author list, and corresponding author—as well as timestamped responses from reviewer invitees. The dataset also includes article-level metadata detailing editorial decisions, DOIs, and structured author rosters identifying the corresponding author for each submission.

S1.2 Publication Data

We use SciSciNet database [1] and the Web of Science (WoS) database to extract individual publication records and citation behaviors. SciSciNet is a large-scale open-access data lake for science of science research, encompassing over 134 million scientific publications and extensive metadata on funding, public engagement, and knowledge diffusion until 2022. To track how reviewer invitees' citation behaviors evolve pre- and post-invitation, and to control for person- and paper-level covariates, we integrated six SciSciNet datasets:

- SciSciNet_Papers, which records every primary paper together with its unique ID, disciplinary category, entry counts, and foundational metrics.
- SciSciNet_Authors_Gender, which assigns each scholar a unique ID, name, and career-level indicators.
- SciSciNet_PaperAuthorAffiliations, which traces every linkage between papers, authors, and their affiliations.
- SciSciNet_PaperReferences, which captures every citing–cited pair among the primary papers.
- SciSciNet_Affiliations, which maps affiliation IDs to institution names and summary metrics.
- SciSciNet_Journals, which details journal IDs alongside titles, ISSNs, publishers, and official webpages.

- SciSciNet_PaperFields, which lists field IDs, names, and flags that indicate whether each field is top-level or a subdiscipline.

In addition, we use the WoS database to extract the citation position within each paper’s reference list. This allows us to compute the prominence metric shown in Figure 4, which assesses the conceptual centrality of the corresponding author’s work within the reviewer invitees’ research. Specifically, we use the reciprocal of the earliest citation position as a proxy for prominence: references cited earlier in a list are assumed to be more central to the citing paper’s argument.

S1.3 Record Linkage and Panel Assembly

We first linked invitee and author identities in the peer review dataset to unique scholar identifiers in SciSciNet. To maximize accuracy, we followed a multi-step matching procedure: (1) matching based on first initial and last name; (2) restricting candidates to SciSciNet scholars with at least three publications to ensure a robust publication record; (3) using fuzzy string matching to compare full names and affiliations, generating similarity scores for each potential match; and (4) prioritizing candidates who had previously published in the same journal or were located close to the journal in the citation network. To validate match quality, we compared corresponding authors’ and reviewers’ email addresses from the journal’s editorial system with researchers’ emails listed in publications, assuming email addresses uniquely identify individuals. This approach yielded an estimated match accuracy of over 83%.

Each manuscript was linked to the SciSciNet ID of its corresponding author, forming dyadic reviewer–submission pairs. The resulting dataset comprised 550,686 invitations tied to 37,279 unique submissions and 209,122 distinct matched reviewer invitees. Reviewer invitees’ responses were then numerically encoded into three categories: Accept (clicked “Agreed”), Click-Denial (clicked “Decline” or marked “Unavailable”), and Auto-Denial (no response).

To recover local time and weekday context for each invitation, we converted the original timestamps to the reviewer invitees’ local time zone—using affiliation-based location data when available or inferring time zones from the country code in the email domain, mapped to IANA standards.

This procedure yielded a longitudinal panel of 5.74 million reviewer-year observations, spanning an event-time window from 10 years prior to 5 years following the invitation. Each row in this panel includes six citation-based outcome measures (described in Figure 4), covariates, treatment indicators based on reviewer invitees' behavior, and fixed identifiers for the reviewer invitees, manuscript, and calendar year.

S2 Descriptive Summary Between Treatment and Control Groups

In our analysis, we adjust for three categories of factors that may influence post-invitation citation behavior. First, to account for potential pre-existing ties between reviewer invitees and the manuscript's corresponding author, we construct three measures of prior relationship strength: (1) the number of co-authored papers between the reviewer invitees and the corresponding author prior to the invitation; (2) the cumulative number of times the reviewer invitees had cited the corresponding author's work; and (3) the cumulative number of times the corresponding author had cited the reviewer invitee's work. Second, to capture reviewer invitees' research experience and scholarly impact, we compute each reviewer invitees' career *h*-index (log) and total publication count (log) as of the invitation date. These measures allow us to assess whether more established researchers systematically differ in their responsiveness or exposure-related behaviors. Third, to proxy the reviewer invitees' relationship with the journal, we calculate two variables: (1) the number of publications by the reviewer invitees in the focal journal prior to the invitation, and (2) the number of times the reviewer invitees cited articles from the focal journal before the invitation. These metrics reflect prior intellectual engagement and visibility within the journal's ecosystem. These citation-based links provide a fine-grained measure of scholarly proximity that could confound post-invitation citation outcomes.

These covariates are included to ensure that any estimated differences between the Click-Delay and Auto-Delay groups are not simply attributable to underlying differences in expertise, institutional alignment, or pre-existing author-reviewer relationships.

TABLE S1: SUMMARY STATISTICS OF CONTROL AND TREATMENT GROUPS

Before Invitation (N = 450,737)					
	Click-Delay Group (N=246,565)		Auto-Delay Group (N=204,172)		T-Test
	Mean	Std.	Mean	Std.	t-stat
# Collaboration for Both	0.03	0.65	0.04	2.43	-1.50
# Citation Reviewer to Author	0.57	5.98	0.55	8.93	0.62
# Citation Author to Reviewer	0.95	9.10	0.89	10.11	1.91*
Reviewer # Publications (log)	4.40	1.23	4.17	1.36	58.84***
Reviewer H-index (log)	2.38	1.09	2.39	1.08	-3.00***
Reviewer # Pub on Journal	1.14	2.78	0.75	2.25	51.43***
Reviewer # Ref to Journal	18.10	30.31	13.34	27.25	55.53***

Table S1. Descriptive statistics of pre-invitation reviewer invitees' characteristics: Click-Delay vs. Auto-Delay groups (two-sample *t*-tests). Variables summarized include (i) reviewer–author proximity—the number of prior co-authored papers plus cumulative citations flowing from reviewer invitees to author and vice versa; (ii) reviewer invitees' research experience—log counts of total publications and H-index; and (iii) engagement with the journal—pre-invitation publications by the reviewer invitees in the focal journal and the number of times the reviewer invitees cited articles from the focal journal. Means and standard deviations are shown for each group, with the final column reporting two-sample *t*-statistics. (***) $p < 0.01$, (**) $p < 0.05$, (*) $p < 0.1$).

S3 Supplementary Results for Difference-in-Difference Results

In the main text, we employed a canonical Difference-in-Differences (DiD) framework and documented a significant divergence in post-invitation citation behavior between Click-Denial reviewer invitees (minimally exposed) and Auto-Denial reviewer invitees (unexposed). To evaluate the robustness of these findings, we complement our primary analysis with additional specifications that address potential heterogeneity in treatment timing and unobserved individual differences, including TWFE and staggered DiD.

S3.1 Two-Way Fixed Effect Difference-in-Differences

We estimate a static two-way fixed-effects (TWFE) difference-in-differences model that absorbs both unit-specific heterogeneity and common calendar-year shocks. Each observation is a reviewer–manuscript pair i observed in year t :

$$y_{it} = \beta \text{ Treat}_i \times \text{Post}_t + \gamma_i + \delta_t + \epsilon_{it}$$

where y_{it} is the citation outcome for unit i in period t ; $\text{Treat}_i \times \text{Post}_t$ is the interaction term indicating treated reviewers in the post-invitation period; γ_i is unit fixed effect; δ_t captures calendar year fixed effects respectively, and ϵ_{it} is the error term. The coefficient β represents the difference-in-differences estimate of the invitation’s effect on citations.

Estimates from this model, presented in Table S2, show results consistent with our main DiD specification. The estimated effect remains statistically significant at 0.0021 citations per year ($p = 0.000$), reflecting a 14% relative increase. Because the model includes individual fixed effects, all time-invariant covariates are absorbed and thus not separately estimated.